

Yu Liu

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EDUCATION

The Chinese University of Hong Kong 2022.08 - 2025.07
Faculty of Medicine, Department of Medicine and Therapeutics, Division of Neurology
Ph.D. in Medical Sciences

Beihang University 2018.09 - 2021.06
Postgraduate recommendation
School of Biological Science and Medical Engineering
M.phil. in Biomedical Engineering, GPA: 3.80/4.00
Core courses: Statistics, Biomechanics, Detection and processing experiment of biomedical signal, Modeling and simulation experiment of biological system, Anatomy and physiology, Ergonomics
Thesis: Hemodynamic study of vertebrobasilar dolichoectasia

Hebei University of Technology 2014.09 - 2018.06
School of Electrical Engineering
B.Eng. in Biomedical Engineering, GPA: 3.71/4.00
Honor Graduate with Distinction, Rank: 1/41
Core courses: Linear algebra, Engineering mathematics, Fundamental Theory of Circuit, Signal and linear system, Automatic control theory, Digital signal processing, Digital image processing, Medical imaging system, Fundamentals of mono-chip computers and applications, Biomedical electronics, Biomedical sensor and measurement
Thesis: Heart rate detection based on photoplethysmographic imaging (**Best thesis award, top 5%**)

WORK EXPERIENCE

Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences 2023.09-2023.11
Visiting Student, Part-time
Cooperation Project: The pathogenic and recurrent mechanisms of symptomatic intracranial atherosclerotic stroke (sICAS): an interdisciplinary study based on advanced neuroimaging and computational fluid dynamics (CFD)

Fosun Aitrox, Shanghai 2021.06-2022.06
Research and Development Engineer, Full-time
Project: Hemodynamic algorithm for fraction flow reserve computed tomography (FFRct)

China National Institute of Standardization, Beijing 2019.08-2019.10
Research Intern, Part-time
Project: Standardized and digital management of medical insurance policy

PUBLICATIONS

Journal

- **More severe cerebral small vessel disease associated with poor leptomeningeal collaterals in symptomatic intracranial atherosclerotic stenosis**

- Yuying Liu, Shuang Li, Xuan Tian, Jill Abrigo, Bonnie YK Lam, Jize Wei, Lina Zheng, **Yu Liu**, Ziqi Li, Tingjun Liang, Bonaventure YM Ip, Thomas W Leung, Xinyi Leng
Journal of Cerebral Blood Flow & Metabolism (JCBFM), 2024 (**Q1**, **IF: 4.9**)
- **Clinical implications of haemodynamics in symptomatic intracranial atherosclerotic stenosis by computational fluid dynamics modelling: a systematic review**
Yu Liu, Shuang Li, Haipeng Liu, Xuan Tian, Yuying Liu, Ziqi Li, Thomas W Leung, Xinyi Leng
Stroke and Vascular Neurology (SVN), 2024 (**Q1**, **IF: 4.4**)
 - **Cerebral hemodynamics and stroke risks in symptomatic intracranial atherosclerotic stenosis with internal versus cortical borderzone infarcts: a computational fluid dynamics study**
Shuang Li, Xuan Tian, Bonaventure Ip, Xueyan Feng, Hing Lung Ip, Jill Abrigo, Linfang Lan, Haipeng Liu, Lina Zheng, Yuying Liu, **Yu Liu**, Karen KY Ma, Florence SY Fan, Sze Ho Ma, Hui Fang, Yuming Xu, Alexander Y Lau, Howan Leung, Yannie OY Soo, Vincent CT Mok, Ka Sing Wong, Xinyi Leng, Thomas W Leung
Journal of Cerebral Blood Flow & Metabolism (JCBFM), 2024 (**Q1**, **IF: 4.9**)
 - **Hemodynamic significance of intracranial atherosclerotic disease and ipsilateral imaging markers of cerebral small vessel disease**
Lina Zheng, Xuan Tian, Jill Abrigo, Hui Fang, Bonaventure YM Ip, Yuying Liu, Shuang Li, **Yu Liu**, Linfang Lan, Haipeng Liu, Hing Lung Ip, Florence SY Fan, Sze Ho Ma, Karen Ma, Alexander Y Lau, Yannie OY Soo, Howan Leung, Vincent CT Mok, Lawrence KS Wong, Yuming Xu, Liping Liu, Xinyi Leng, Thomas W Leung
European Stroke Journal (ESJ), 2024 (**Q1**, **IF: 5.8**)
 - **Effects of abnormal vertebral arteries and the circle of Willis on vertebrobasilar dolichoectasia: a multi-scale simulation study**
Yu Liu, Xinmiao Zhang, Yawei Wang, Wentao Feng, Jing Jing, Zhujun Sun, Bitian Wang, Yongjun Wang, Yubo Fan
Clinical Biomechanics (Clin Biomech), 2023 (**Q3**, **IF: 1.4**)
 - **Effects of stent shape on focal hemodynamics in intracranial atherosclerotic stenosis: A simulation study with computational fluid dynamics modeling**
Haipeng Liu, **Yu Liu**, Bonaventure Y M Ip, Sze Ho Ma, Jill Abrigo, Yannie O Y Soo, Thomas W Leung, Xinyi Leng
Frontiers in Neurology (Front Neurol), 2022 (**Q3**, **IF: 2.7**)
 - **Variations of human cerebral and ocular blood flow during exposure to multi-axial accelerations: A mathematical modeling study**
Weipeng Li, Bitian Wang, Yawei Wang, Xiaoyu Liu, Wentao Feng, Tianya Liu, Zhujun Sun, **Yu Liu**, Songyang Liu, Yubo Fan
Medical & Biological Engineering & Computing (MBEC), 2022 (**Q3**, **IF: 2.6**)

Conference

- **Associations of ICA Bifurcation Geometry, MCA Geometry and Plaque Location in Symptomatic Atherosclerotic MCA Stenosis**
Yu Liu, Shuang Li, Linfang Lan, Xuan Tian, Yuying Liu, Ziqi Li, Haipeng Liu, Bonaventure Ym Ip, Thomas Leung, Xinyi Leng
Poster presentation, European Stroke Organisation Conference (ESOC) 2024
- **Association between Adjacent Arterial Geometry and Plaque Features in Symptomatic Atherosclerotic Middle Cerebral Artery Stenosis**
Yu Liu, Shuang Li, Xuan Tian, Yuying Liu, Ziqi Li, Haipeng Liu, Bonaventure YM Ip, Thomas W Leung, Xinyi Leng
Poster presentation, Asia Pacific Stroke Conference (APSC) 2023
- **Cerebral hemodynamics by computational fluid dynamics modeling in symptomatic intracranial atherosclerotic disease and the clinical relevance: a systematic review**

Yu Liu, Shuang Li, Xuan Tian, Yuying Liu, Haipeng Liu, Thomas Leung, Xinyi Leng
Poster presentation, European Stroke Organisation Conference (ESOC) 2023

PROJECT

- **The pathogenic and recurrent mechanisms of symptomatic intracranial atherosclerotic stroke (sICAS): an interdisciplinary study based on advanced neuroimaging and computational fluid dynamics (CFD)**

National Natural Science Foundation of China (NSFC)/ Research Grants Council (RGC) of Hong Kong Joint Research Scheme, 2021-2023 (Participant)

Affiliated institution: Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences/The Chinese University of Hong Kong

- **Impact of blood pressure variations on focal and global cerebral hemodynamics in symptomatic intracranial atherosclerotic stenosis (sICAS): a pilot study with patient-specific computational fluid dynamics (CFD) modeling**

Health and Medical Research Fund (HMRF), 2018-2022 (Participant)

Affiliated institution: The Chinese University of Hong Kong/ Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences

- **Intelligent Circulatory System Rehabilitation Device Based on Extracorporeal Counterpulsation (ECP) Technology**

National Key Research and Development Plan, Ministry of Science and Technology of the People's Republic of China, 2020-2023 (Participant)

Affiliated institution: Sun Yat-sen University/Beihang University

CHALLENGE

First prize, National Biomedical Engineering Innovation Design Competition for College Students (**top 10%**)

Project: Low-power and remote electrocardiogram (ECG) monitoring system

Third prize, iCAN Innovation Contest

Project: Intelligent wheelchair based on real-time pose transformation

HONORS&AWARDS

Research postgraduate scholarship , the Chinese University of Hong Kong	2022-2025
First-class scholarship , Beihang University (3 times)	2018-2021
Outstanding graduate , Hebei Province (top 5%)	2018.06
Outstanding graduate , Hebei University of Technology	2018.06
First-class scholarship , Hebei University of Technology (2 times)	2017-2018
Outstanding student leader , Hebei University of Technology	2017.12
Outstanding student , Hebei University of Technology	2016.12